

Mycelic

Loss accounting and gap decomposition

QUESTION

How should thousands of private user-level agents abstract, route, combine, question, verify and propagate what they know upward, so that an enterprise kernel discovers what no individual could?

ENTERPRISE SIZES MEASURED

— users

CONTENTS

8 sections · 26 tables · 0 figures

GENERATED

2026-09-22

Every table and figure in this document is computed from the raw per-run records in `research/mycelic/artifacts/` at generation time. No number is transcribed by hand. Section 5 states exactly which results were measured on real models and which were simulated.

What this is. The benchmark report says the hierarchy finds 36% of the hidden patterns at 50,000 users and the strongest centralised system finds 70%. This document says *where the other patterns went*. Every discoverable gold pattern is traced through the pipeline and the first stage at which it is lost is recorded, for the hierarchy (`H_mycelic_full`), its centralised twin (`B4_central_triage`), the perfect-retrieval oracle (`Y_oracle_retrieval`) and the strongest centralised baseline (`A2_chunked_ctx`), on the same worlds. Then three paired diagnostics on identical worlds decompose the gap into the stages responsible.

Nothing here is a design. It is the measurement the next design is built on. Every number is computed from

`research/mycelic/artifacts/loss_funnel.jsonl` and `quick_diag*.jsonl` at generation time. 5 seeds at 10,000 users, 3 at 50,000.

1. The finding in one paragraph

The handoff pack's leading hypothesis was that the sketch channel is too poor to seed good hypotheses. It is not: at 10,000 users the sketch triage sees 98.5% of gold anchors. The pattern dies later, at two places. First, the **question budget**: the calibrated budget asks only 63.5% of gold anchors at 10k and 43% at 50k, and that is the single largest first-loss stage at both scales (47% and 51% of the gap to A2). Second, the **register**: at 10k, 12% of all patterns are matched at confidence ≥ 0.5 and then cut, because the hierarchy rates spurious candidates as highly as genuine ones and the register holds one entry per entity. A2 loses nothing at the register. Asking every triage candidate recovers evidence for 98.5% of patterns and the kernel forms a correct confident candidate for ~80% — more than A2 — and then buries them. The calibrated small budget was the right choice *given the ranker*; the ranker is the fault. At 50,000 users the sketch additionally loses 18% of patterns (mostly rare: a single-witness facet cannot set a site bit under the support threshold), so scale adds a third, smaller stage.

2. Stage-by-stage survival

Stages are ordered by pipeline position. They are not strictly nested — a pattern can be visible to the sketch without having been extracted — so `first stage lost` records the first stage in order that failed. The flat systems have no sketch, triage, question or descent stage.

10,000 users (5 seeds)

Stage survival, all patterns

stage	A2_chunked_ctx	B4_central_triage	H_mycelic_full	Y_oracle_retrieval
extracted (≥ 2 facets, right entity+predicate)	1.000 [1.00, 1.00]	0.970 [0.95, 0.99]	0.920 [0.89, 0.96]	1.000 [1.00, 1.00]
visible to sketch triage (≥ 2 foreign sites, ≥ 2 regions, span ≥ 2)	—	—	0.985 [0.97, 0.99]	—
on the triage candidate list	—	—	0.985 [0.97, 0.99]	—
inside the question budget	—	—	0.635 [0.58, 0.70]	—
descent reached a facet holder	—	—	0.665 [0.61, 0.73]	—
≥ 2 gold links in kernel pool (= evidence coverage)	1.000 [1.00, 1.00]	0.970 [0.95, 0.99]	0.690 [0.64, 0.74]	1.000 [1.00, 1.00]
candidate formed on right entity + chain	0.685 [0.62, 0.73]	0.385 [0.33, 0.44]	0.595 [0.54, 0.65]	0.250 [0.21, 0.28]
matched primary rule at ANY confidence	0.685 [0.62, 0.74]	0.370 [0.32, 0.42]	0.540 [0.45, 0.63]	0.240 [0.19, 0.28]
matched with confidence ≥ 0.5	0.685 [0.62, 0.74]	0.370 [0.32, 0.42]	0.515 [0.44, 0.61]	0.240 [0.19, 0.28]
survived the register cut (= reported)	0.685 [0.62, 0.74]	0.370 [0.32, 0.42]	0.375 [0.32, 0.43]	0.240 [0.19, 0.28]

patterns counted: A2_chunked_ctx: 200, B4_central_triage: 200, H_mycelic_full: 200, Y_oracle_retrieval: 200

Stage survival, rare-signal patterns only

stage	A2_chunked_ctx	B4_central_triage	H_mycelic_full	Y_oracle_retrieval
extracted (≥ 2 facets, right entity+predicate)	1.000 [1.00, 1.00]	0.955 [0.92, 0.99]	0.815 [0.75, 0.88]	1.000 [1.00, 1.00]
visible to sketch triage (≥ 2 foreign sites, ≥ 2 regions, span ≥ 2)	—	—	0.977 [0.95, 1.00]	—

stage	A2_chunked_ctx	B4_central_triage	H_mycelic_full	Y_oracle_retrieval
on the triage candidate list	—	—	0.977 [0.95, 1.00]	—
inside the question budget	—	—	0.446 [0.36, 0.54]	—
descent reached a facet holder	—	—	0.463 [0.41, 0.53]	—
>=2 gold links in kernel pool (= evidence coverage)	1.000 [1.00, 1.00]	0.955 [0.92, 0.99]	0.500 [0.42, 0.58]	1.000 [1.00, 1.00]
candidate formed on right entity + chain	0.653 [0.55, 0.75]	0.258 [0.19, 0.33]	0.416 [0.30, 0.54]	0.143 [0.10, 0.18]
matched primary rule at ANY confidence	0.653 [0.55, 0.75]	0.258 [0.19, 0.33]	0.360 [0.26, 0.46]	0.143 [0.10, 0.18]
matched with confidence >= 0.5	0.653 [0.55, 0.75]	0.258 [0.19, 0.33]	0.341 [0.25, 0.45]	0.143 [0.10, 0.18]
survived the register cut (= reported)	0.653 [0.55, 0.75]	0.258 [0.19, 0.33]	0.215 [0.16, 0.27]	0.143 [0.10, 0.18]

patterns counted: A2_chunked_ctx: 78, B4_central_triage: 78, H_mycelic_full: 78, Y_oracle_retrieval: 78

Stage survival, common patterns only

stage	A2_chunked_ctx	B4_central_triage	H_mycelic_full	Y_oracle_retrieval
extracted (>=2 facets, right entity+predicate)	1.000 [1.00, 1.00]	0.986 [0.96, 1.00]	0.992 [0.98, 1.00]	1.000 [1.00, 1.00]
visible to sketch triage (>=2 foreign sites, >=2 regions, span>=2)	—	—	0.993 [0.98, 1.00]	—
on the triage candidate list	—	—	0.993 [0.98, 1.00]	—
inside the question budget	—	—	0.771 [0.66, 0.85]	—
descent reached a facet holder	—	—	0.798 [0.72, 0.88]	—
>=2 gold links in kernel pool (= evidence coverage)	1.000 [1.00, 1.00]	0.986 [0.96, 1.00]	0.821 [0.76, 0.88]	1.000 [1.00, 1.00]
candidate formed on right entity + chain	0.715 [0.68, 0.76]	0.475 [0.35, 0.58]	0.721 [0.69, 0.76]	0.325 [0.27, 0.37]
matched primary rule at ANY confidence	0.716 [0.67, 0.76]	0.450 [0.34, 0.55]	0.660 [0.56, 0.75]	0.307 [0.25, 0.35]
matched with confidence >= 0.5	0.716 [0.67, 0.76]	0.450 [0.34, 0.55]	0.630 [0.54, 0.72]	0.307 [0.25, 0.35]
survived the register cut (= reported)	0.716 [0.67, 0.76]	0.450 [0.34, 0.55]	0.484 [0.43, 0.55]	0.307 [0.25, 0.35]

patterns counted: A2_chunked_ctx: 122, B4_central_triage: 122, H_mycelic_full: 122, Y_oracle_retrieval: 122

Where the hierarchy loses each pattern FIRST

first stage lost	all	rare	common
extracted (>=2 facets, right entity+predicate)	16 (8%)	15	1
visible to sketch triage (>=2 foreign sites, >=2 regions, span>=2)	3 (2%)	2	1
on the triage candidate list	0 (0%)	0	0
inside the question budget	62 (31%)	34	28
descent reached a facet holder	1 (0%)	0	1
>=2 gold links in kernel pool (= evidence coverage)	0 (0%)	0	0
candidate formed on right entity + chain	17 (8%)	5	12
matched primary rule at ANY confidence	11 (6%)	4	7
matched with confidence >= 0.5	4 (2%)	0	4
survived the register cut (= reported)	24 (12%)	8	16
reported (survived every stage)	62 (31%)	10	52
total	200	78	122

Why the sketch could not see the invisible ones

why the sketch could not see it	all	rare	common	median witnesses
causal span < 2 across foreign sites	3	2	1	7

Discovery by number of witnesses (hierarchy)

witnesses (records)	n	visible to sketch	in pool	reported
0–4	12	1.00	0.75	0.25
5–7	52	0.96	0.46	0.21
8–11	16	1.00	0.44	0.12
12–16	13	1.00	0.62	0.15
17–25	65	0.98	0.78	0.45
26–∞	42	1.00	0.93	0.67

Discovery by number of witnesses (A2_chunked_ctx)

witnesses (records)	n	visible to sketch	in pool	reported
0–4	12	—	1.00	0.67
5–7	52	—	1.00	0.56
8–11	16	—	1.00	0.81
12–16	13	—	1.00	0.77
17–25	65	—	1.00	0.74
26–∞	42	—	1.00	0.69

Where the reported patterns sit in the register

architecture	reported	median rank	within top 40	within top 100	median confidence
A2_chunked_ctx	137	197	19	37	0.85
B4_central_triage	74	263	9	16	0.88
H_mycelic_full	75	255	8	18	0.89
Y_oracle_retrieval	48	279	3	9	0.91

Decomposing the gap to the strongest centralised system

Paired on 200 (seed, pattern) pairs at 10,000 users: A2_chunked_ctx reports 137 (68.5%), H_mycelic_full reports 75 (37.5%). A2_chunked_ctx finds 77 that H_mycelic_full misses; H_mycelic_full finds 15 that A2_chunked_ctx misses.

Of the 77 patterns A2_chunked_ctx reports and H_mycelic_full does not, the first stage H_mycelic_full lost them at:

stage	patterns	share of the gap	of which rare
extracted (>=2 facets, right entity+predicate)	3	4%	3
visible to sketch triage (>=2 foreign sites, >=2 regions, span>=2)	1	1%	1
inside the question budget	36	47%	22
candidate formed on right entity + chain	6	8%	2
matched primary rule at ANY confidence	7	9%	3
matched with confidence >= 0.5	3	4%	0
survived the register cut (= reported)	21	27%	7

Decomposing the gap to the centralised twin (same algorithm)

Paired on 200 (seed, pattern) pairs at 10,000 users: B4_central_triage reports 74 (37.0%), H_mycelic_full reports 75 (37.5%). B4_central_triage finds 34 that H_mycelic_full misses; H_mycelic_full finds 35 that B4_central_triage misses.

Of the 34 patterns B4_central_triage reports and H_mycelic_full does not, the first stage H_mycelic_full lost them at:

stage	patterns	share of the gap	of which rare
inside the question budget	18	53%	7
descent reached a facet holder	1	3%	0
candidate formed on right entity + chain	4	12%	1
matched primary rule at ANY confidence	2	6%	1
matched with confidence ≥ 0.5	2	6%	0
survived the register cut (= reported)	7	21%	1

Decomposing the gap to the perfect-retrieval oracle

Paired on 200 (seed, pattern) pairs at 10,000 users: Y_oracle_retrieval reports 48 (24.0%), H_mycelic_full reports 75 (37.5%). Y_oracle_retrieval finds 16 that H_mycelic_full misses; H_mycelic_full finds 43 that Y_oracle_retrieval misses.

Of the 16 patterns Y_oracle_retrieval reports and H_mycelic_full does not, the first stage H_mycelic_full lost them at:

stage	patterns	share of the gap	of which rare
inside the question budget	11	69%	6
candidate formed on right entity + chain	1	6%	0
matched primary rule at ANY confidence	2	12%	1
matched with confidence ≥ 0.5	1	6%	0
survived the register cut (= reported)	1	6%	0

50,000 users (3 seeds)

Stage survival, all patterns

stage	A2_chunked_ctx	B4_central_triage	H_mycelic_full	Y_oracle_retrieval
extracted (≥ 2 facets, right entity+predicate)	1.000 [1.00, 1.00]	0.493 [0.47, 0.51]	0.900 [0.87, 0.94]	1.000 [1.00, 1.00]
visible to sketch triage (≥ 2 foreign sites, ≥ 2 regions, span ≥ 2)	—	—	0.780 [0.74, 0.80]	—
on the triage candidate list	—	—	0.780 [0.74, 0.80]	—
inside the question budget	—	—	0.427 [0.36, 0.47]	—
descent reached a facet holder	—	—	0.427 [0.35, 0.48]	—
≥ 2 gold links in kernel pool (= evidence coverage)	1.000 [1.00, 1.00]	0.493 [0.47, 0.51]	0.450 [0.38, 0.49]	1.000 [1.00, 1.00]
candidate formed on right entity + chain	0.727 [0.70, 0.76]	0.380 [0.34, 0.41]	0.397 [0.34, 0.44]	0.280 [0.16, 0.35]
matched primary rule at ANY confidence	0.703 [0.68, 0.74]	0.323 [0.31, 0.34]	0.363 [0.31, 0.39]	0.267 [0.16, 0.34]
matched with confidence ≥ 0.5	0.703 [0.68, 0.74]	0.323 [0.31, 0.34]	0.363 [0.31, 0.39]	0.267 [0.16, 0.34]
survived the register cut (= reported)	0.703 [0.68, 0.74]	0.323 [0.31, 0.34]	0.363 [0.31, 0.39]	0.267 [0.16, 0.34]

patterns counted: A2_chunked_ctx: 300, B4_central_triage: 300, H_mycelic_full: 300, Y_oracle_retrieval: 300

Stage survival, rare-signal patterns only

stage	A2_chunked_ctx	B4_central_triage	H_mycelic_full	Y_oracle_retrieval
extracted (>=2 facets, right entity+predicate)	1.000 [1.00, 1.00]	0.213 [0.18, 0.24]	0.756 [0.69, 0.82]	1.000 [1.00, 1.00]
visible to sketch triage (>=2 foreign sites, >=2 regions, span>=2)	—	—	0.558 [0.44, 0.67]	—
on the triage candidate list	—	—	0.558 [0.44, 0.67]	—
inside the question budget	—	—	0.307 [0.23, 0.36]	—
descent reached a facet holder	—	—	0.274 [0.15, 0.36]	—
>=2 gold links in kernel pool (= evidence coverage)	1.000 [1.00, 1.00]	0.213 [0.18, 0.24]	0.314 [0.23, 0.36]	1.000 [1.00, 1.00]
candidate formed on right entity + chain	0.639 [0.62, 0.68]	0.178 [0.13, 0.25]	0.263 [0.18, 0.32]	0.238 [0.11, 0.32]
matched primary rule at ANY confidence	0.622 [0.56, 0.68]	0.157 [0.10, 0.21]	0.235 [0.15, 0.29]	0.230 [0.11, 0.32]
matched with confidence >= 0.5	0.622 [0.56, 0.68]	0.157 [0.10, 0.21]	0.235 [0.15, 0.29]	0.230 [0.11, 0.32]
survived the register cut (= reported)	0.622 [0.56, 0.68]	0.157 [0.10, 0.21]	0.235 [0.15, 0.29]	0.230 [0.11, 0.32]

patterns counted: A2_chunked_ctx: 112, B4_central_triage: 112, H_mycelic_full: 112, Y_oracle_retrieval: 112

Stage survival, common patterns only

stage	A2_chunked_ctx	B4_central_triage	H_mycelic_full	Y_oracle_retrieval
extracted (>=2 facets, right entity+predicate)	1.000 [1.00, 1.00]	0.662 [0.61, 0.72]	0.990 [0.98, 1.00]	1.000 [1.00, 1.00]
visible to sketch triage (>=2 foreign sites, >=2 regions, span>=2)	—	—	0.911 [0.89, 0.93]	—
on the triage candidate list	—	—	0.911 [0.89, 0.93]	—
inside the question budget	—	—	0.501 [0.44, 0.55]	—
descent reached a facet holder	—	—	0.522 [0.48, 0.56]	—
>=2 gold links in kernel pool (= evidence coverage)	1.000 [1.00, 1.00]	0.662 [0.61, 0.72]	0.533 [0.48, 0.58]	1.000 [1.00, 1.00]
candidate formed on right entity + chain	0.781 [0.76, 0.79]	0.509 [0.44, 0.59]	0.479 [0.44, 0.51]	0.307 [0.20, 0.36]
matched primary rule at ANY confidence	0.754 [0.75, 0.76]	0.430 [0.36, 0.49]	0.444 [0.41, 0.49]	0.292 [0.20, 0.35]
matched with confidence >= 0.5	0.754 [0.75, 0.76]	0.430 [0.36, 0.49]	0.444 [0.41, 0.49]	0.292 [0.20, 0.35]
survived the register cut (= reported)	0.754 [0.75, 0.76]	0.430 [0.36, 0.49]	0.444 [0.41, 0.49]	0.292 [0.20, 0.35]

patterns counted: A2_chunked_ctx: 188, B4_central_triage: 188, H_mycelic_full: 188, Y_oracle_retrieval: 188

Where the hierarchy loses each pattern FIRST

first stage lost	all	rare	common
extracted (>=2 facets, right entity+predicate)	30 (10%)	28	2
visible to sketch triage (>=2 foreign sites, >=2 regions, span>=2)	54 (18%)	39	15
on the triage candidate list	0 (0%)	0	0
inside the question budget	100 (33%)	23	77
descent reached a facet holder	1 (0%)	1	0
>=2 gold links in kernel pool (= evidence coverage)	0 (0%)	0	0
candidate formed on right entity + chain	13 (4%)	4	9
matched primary rule at ANY confidence	7 (2%)	1	6
matched with confidence >= 0.5	0 (0%)	0	0
survived the register cut (= reported)	0 (0%)	0	0
reported (survived every stage)	95 (32%)	16	79

first stage lost	all	rare	common
total	300	112	188

Why the sketch could not see the invisible ones

why the sketch could not see it	all	rare	common	median witnesses
fewer than 2 foreign sites	15	10	5	5
foreign sites in fewer than 2 regions	8	7	1	7
causal span < 2 across foreign sites	43	32	11	7

Discovery by number of witnesses (hierarchy)

witnesses (records)	n	visible to sketch	in pool	reported
0–4	16	0.56	0.44	0.31
5–7	75	0.55	0.28	0.20
8–11	22	0.59	0.32	0.27
12–16	25	0.80	0.32	0.20
17–25	104	0.91	0.49	0.41
26–∞	58	0.97	0.71	0.60

Discovery by number of witnesses (A2_chunked_ctx)

witnesses (records)	n	visible to sketch	in pool	reported
0–4	16	—	1.00	0.44
5–7	75	—	1.00	0.61
8–11	22	—	1.00	0.77
12–16	25	—	1.00	0.84
17–25	104	—	1.00	0.70
26–∞	58	—	1.00	0.81

Where the reported patterns sit in the register

architecture	reported	median rank	within top 40	within top 100	median confidence
A2_chunked_ctx	211	851	17	34	0.85
B4_central_triage	97	667	3	9	0.84
H_mycelic_full	109	886	1	5	0.88
Y_oracle_retrieval	80	1268	1	6	0.92

Decomposing the gap to the strongest centralised system

Paired on 300 (seed, pattern) pairs at 50,000 users: A2_chunked_ctx reports 211 (70.3%), H_mycelic_full reports 109 (36.3%). A2_chunked_ctx finds 125 that H_mycelic_full misses; H_mycelic_full finds 23 that A2_chunked_ctx misses.

Of the 125 patterns A2_chunked_ctx reports and H_mycelic_full does not, the first stage H_mycelic_full lost them at:

stage	patterns	share of the gap	of which rare
extracted (>=2 facets, right entity+predicate)	13	10%	11
visible to sketch triage (>=2 foreign sites, >=2 regions, span>=2)	38	30%	26
inside the question budget	64	51%	13
candidate formed on right entity + chain	5	4%	1
matched primary rule at ANY confidence	5	4%	0

Decomposing the gap to the centralised twin (same algorithm)

Paired on 300 (seed, pattern) pairs at 50,000 users: B4_central_triage reports 97 (32.3%), H_mycelic_full reports 109 (36.3%). B4_central_triage finds 32 that H_mycelic_full misses; H_mycelic_full finds 44 that B4_central_triage misses.

Of the 32 patterns B4_central_triage reports and H_mycelic_full does not, the first stage H_mycelic_full lost them at:

stage	patterns	share of the gap	of which rare
extracted (≥ 2 facets, right entity+predicate)	1	3%	1
inside the question budget	27	84%	1
candidate formed on right entity + chain	2	6%	0
matched primary rule at ANY confidence	2	6%	0

Decomposing the gap to the perfect-retrieval oracle

Paired on 300 (seed, pattern) pairs at 50,000 users: Y_oracle_retrieval reports 80 (26.7%), H_mycelic_full reports 109 (36.3%). Y_oracle_retrieval finds 46 that H_mycelic_full misses; H_mycelic_full finds 75 that Y_oracle_retrieval misses.

Of the 46 patterns Y_oracle_retrieval reports and H_mycelic_full does not, the first stage H_mycelic_full lost them at:

stage	patterns	share of the gap	of which rare
extracted (≥ 2 facets, right entity+predicate)	3	7%	3
visible to sketch triage (≥ 2 foreign sites, ≥ 2 regions, span ≥ 2)	15	33%	9
inside the question budget	22	48%	3
candidate formed on right entity + chain	3	7%	0
matched primary rule at ANY confidence	3	7%	0

3. Paired decomposition on identical worlds

Each row is the calibrated hierarchy with one setting changed, run on the same (scale, seed) worlds as the base. **An unbounded register is a benchmark change, not a design**; it appears here only to measure how much discovery is being lost to ranking rather than to retrieval or judgement. Arrows mark a 95% bootstrap interval entirely on one side of zero.

10,000 users

variant	found	evidence cov.	rare recall	AP	FDR	compute	calls
H_mycelic_full (calibrated)	0.375	0.690	0.215	0.0227	0.975	1.09e+06	9.10e+04
ask every triage candidate (question_frac = 1.0) (5 seeds)	0.300	0.985 ↑	0.271	0.0113 ↓	0.980	2.14e+06 ↑	2.21e+05 ↑
unbounded register (diagnostic only) (5 seeds)	0.540 ↑	0.690	0.341 ↑	0.0261 ↑	0.985 ↑	1.16e+06 ↑	9.10e+04
both (5 seeds)	0.795 ↑	0.985 ↑	0.697 ↑	0.0193	0.989 ↑	2.36e+06 ↑	2.21e+05 ↑
full budget, home-site evidence allowed (5 seeds)	0.360	0.985 ↑	0.274	0.0209	0.976	2.76e+06 ↑	2.56e+05 ↑

50,000 users

variant	found	evidence cov.	rare recall	AP	FDR	compute	calls
H_mycelic_full (calibrated)	0.363	0.450	0.235	0.0081	0.987	2.97e+06	2.89e+05
ask every triage candidate (question_frac = 1.0) (3 seeds)	0.407 ↑	0.787 ↑	0.248	0.0085	0.986 ↓	6.27e+06 ↑	6.99e+05 ↑
unbounded register (diagnostic only) (3 seeds)	0.363	0.450	0.235	0.0081	0.987	2.98e+06 ↑	2.89e+05
both (3 seeds)	0.643 ↑	0.787 ↑	0.417 ↑	0.0112 ↑	0.993 ↑	6.83e+06 ↑	6.99e+05 ↑

Reading it. Asking every triage candidate raises evidence coverage from 0.69 to 0.985 at 10k (0.45 to 0.79 at 50k) and *lowers* discovery at 10k: the recovered evidence produces candidates the register cannot hold. Remove the register pressure and the same runs report 0.795 at 10k — above A2's 0.685 — and 0.643 at 50k against A2's 0.703, at two-thirds of A2's compute. Allowing home-site evidence back in does not help, so the flood is not the foreign-only filter. The residual at 50k is the sketch loss, which no question budget can recover because the anchor never reaches the triage list.

4. Why the hierarchy's ranker is worse than A2's on the same operator

Both systems rank with the same confidence logistic. Its main terms — a causal span, several regions, links dominated by different branches — are exactly the properties the triage *pre-selects* anchors for. Among the candidates the hierarchy's kernel actually has to order, those terms carry almost no information, because every questioned anchor already has them. A2's pool holds every anchor, including thousands with no cross-organisational structure at all, for which the same terms are decisive. The hierarchy's discrimination problem is a conditioning problem, not an operator problem.

The consequence is testable: features the triage did **not** condition on — per-link independent support, inter-link lags, conflict volume, dispersion, whether the candidate came from a question — should separate genuine from spurious candidates among the triage survivors. Section 6 measures that.

5. Two smaller losses, measured

Extraction (10% at 50k, 24% of rare patterns). This is stochastic recall, not a capability ceiling: a fresh edge-tier re-read of the same records recovers 80–91% of the patterns lost there, a kernel-tier *local* re-read 100%. A user agent asked a targeted question about an entity can re-read its own notes about it — the entity name is in every record's surface text — so the loss is recoverable without moving any text.

Sketch (18% at 50k, 39 of 54 rare). 43 of the 54 fail on "causal span < 2 across foreign sites": a facet with one witness at a site sets no bit under `sketch_min_support = 2`. Lowering the threshold lets benign single mentions set bits and lengthens the triage list, which only pays if the ranker can absorb the noise — which is the same lever as section 4.

6. How separable are genuine candidates, on kernel-side features?

Every candidate the synthesis produced (before the register cut) was dumped with its confidence-logistic inputs and labelled against gold. One shared logistic was fitted on the pooled candidates of all four systems on the calibration seeds (500–502) and evaluated on the evaluation seeds (0–4) by re-ranking each run's candidates under its own register cap. No feature reads raw text; the privacy accounting is unchanged.

Dump default budget — 10,074 training candidates (seeds 500–502), 16,758 evaluation candidates (seeds 0–4), positive rate 2.4%. Pooled AUC gold-vs-spurious: hand-set confidence 0.609, learned 0.768.

architecture	gold patterns	with any candidate	found (current ranker)	found (learned rank, same #kept)	found (learned, p≥0.5)	AUC current	AUC learned
A2_chunked_ctx	200	137	0.685	0.685	0.460	0.630	0.697
B4_central_triage	200	74	0.370	0.370	0.195	0.562	0.757
H_mycelic_full	200	108	0.375	0.450	0.325	0.692	0.802
Y_oracle_retrieval	200	48	0.240	0.240	0.175	0.521	0.726

Largest standardised weights: `dispersion` -1.53, `verified` +1.41, `n_sites` +1.35, `tspan` -0.80, `min_sup` -0.42, `n_indep` -0.38, `log_n_kos` +0.35.

Dump_qf1 — 9,566 training candidates (seeds 500–502), 16,454 evaluation candidates (seeds 0–4), positive rate 1.0%. Pooled AUC gold-vs-spurious: hand-set confidence 0.672, learned 0.829.

architecture	gold patterns	with any candidate	found (current ranker)	found (learned rank, same #kept)	found (learned, p≥0.5)	AUC current	AUC learned
H_mycelic_full	200	159	0.300	0.550	0.525	0.672	0.829

Largest standardised weights: `verified` +1.37, `dispersion` -1.28, `n_sites` +0.84, `tspan` -0.78, `n_links` +0.77, `min_lag` -0.50, `log_n_kos` -0.26.

7. What this changes about the next architecture

Not "richer sketches". In order of the loss they address:

1. **A calibrated ranker** over kernel-side features, fitted on held-out seeds by the same protocol as every other knob, so the full question budget can be spent without flooding the register. Targets the register loss (12% at 10k; binding at 50k once the budget is raised).
2. **A cheaper descent**, so the full budget is affordable at 50k (it currently doubles compute and calls). Targets the question-budget loss.
3. **Targeted local re-extraction** on descent. Targets the extraction loss.
4. **A support-1 sketch bit** for rare facets, *if* (1) absorbs the extra triage noise. Targets the sketch loss at scale.

Each is a paired experiment against `H_mycelic_full` on identical worlds, with the centralised controls re-run at the same time. None moves raw text.

8. Reproducing this

```
python3 -m research.mycelic.loss_accounting      # 10k x 5, 50k x 3 -> loss_funnel.jsonl
python3 -m research.mycelic.loss_report          # the tables
python3 -m research.mycelic.quick_paired --tag diag_qf1 --cfg '{"question_frac": 1.0}' --seeds 0,1,2,3,4
python3 -m research.mycelic.calibrator dump && python3 -m research.mycelic.calibrator fit
python3 -m research.mycelic.loss_doc             # this document
python3 -m research.mycelic.make_pdf docs/MYCELIC_LOSS_ACCOUNTING.md docs/MYCELIC_LOSS_ACCOUNTING.pdf
```

Generated 2026-09-22 by `research/mycelic/loss_doc.py` from the raw artifacts.